1. Determine whether these matrices are in the reduced row echelon form. Also, find the RRE form of the matrix if it is not in the form.

1.1. $\begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}$

1.2. $\begin{bmatrix} 1 \end{bmatrix}$

1.3. $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

1.4. $\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 2 \end{bmatrix}$

1.5. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

1.6. $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

2. Find the reduced row echelon form of these matrices.

2.1. $\mathbf{A}_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 5 & 7 & 9 \end{bmatrix}$

2.2. $\mathbf{A}_2 = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$

2.3. $\mathbf{A}_3$ is a matrix representing a linear transformation that maps (w, x, y, z) in $\mathbf{R}^4$ to $(w - 2x, x - 2y, y - 2z)$ in $\mathbf{R}^3$

3. Every 3×3 matrix in RRE form falls into 1 of 8 different classes. The **zero** and **identity** matrices which shown below are two of them. What are the other six? (Use a and b to represent arbitrary numbers.)

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad \mathbf{I} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4. Verify that the coefficient matrix of the system below has RRE form and find its solutions.

$$\begin{aligned} x_1 - 2x_3 + 3x_4 &= 0 \\ x_2 + 2x_5 &= 0 \end{aligned}$$

5. Verify that an augmented matrix representing the system of equations below is in RRE form. Also, find a particular solution of the system.

$$\begin{aligned} x_2 + 2x_3 + 3x_5 &= 4 \\ x_4 + 5x_5 &= 6 \end{aligned}$$

6. Find all solutions of the system below.

$$x_1 + 2x_2 + 3x_3 + 4x_4 = 18$$
$$2x_1 + 3x_2 + 4x_3 + x_4 = 24$$

7. For each linear mapping below, find the matrix representing it, and use it to decide if the mapping is one-to-one and/or onto. Also, answer that if the mapping is isomorphism.
(Note that it is possible for a linear mapping that is neither one-to-one nor onto.)

7.1. $S\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^4$ given by $S(x, y) = (x + y, x - y, 2x - y, x + 2y)$

7.2. $T\colon \mathbf{R}^4 \Rightarrow \mathbf{R}^3$ given by $T(w, x, y, z) = (w + 2x + 3y + z, 5x + 7y + 8z, -2w + 11x + 15y + 22z)$

7.3. $U\colon \mathbf{R}^3 \Rightarrow \mathbf{R}^2$ given by $U(x, y, z) = (x + 2y + 3z, 2x + 3y + 4z)$

8. (Challenge) Given a matrix $\mathbf{A}$ and a vector $\mathbf{b} \in \mathbf{R}^m$. If $\begin{bmatrix} 3 \\ -1 \\ 2 \\ 5 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}$, and $\begin{bmatrix} 7 \\ -7 \\ 0 \\ 1 \end{bmatrix}$ all solve the equation $\mathbf{Ax} = \mathbf{b}$.

   Find three solutions of the homogeneous system $\mathbf{Ax} = \mathbf{0}$ and three additional solutions of $\mathbf{Ax} = \mathbf{b}$.